Heuristic Superfluid Cosmology: Collapse-Threshold Estimates for Early-Galaxy Abundance and a Topological Analogue of Collisionless Dark Matter

Amir Benjamin Amitay[1]

[1] *Independent Research*

(Dated: March 21, 2026)

We explore a heuristic cosmological scenario in which the dark sector is modeled as a viscoelastic superfluid condensate whose inertial mass obeys $m = C\rho_0 V$ ($C = 3.523 \pm 0.001$ on 256^3 lattices). Acoustic radiation plus absence of free-streaming lowers the spherical-collapse threshold to an estimated $\delta_c \approx 1.15$. The modified Press-Schechter cumulative halo density then yields a $\sim 6\times$ enhancement at $z = 10$ relative to ΛCDM:

$$\frac{n_{\rm UHF}(>M)}{n_{\Lambda\rm CDM}(>M)} = \frac{\mathrm{erfc}(\delta_{\rm UHF}/\sqrt{2}\sigma)}{\mathrm{erfc}(\delta_{\rm GR}/\sqrt{2}\sigma)} \approx 6.01$$

(with $\sigma = 0.71$), consistent in sign with excess early-galaxy counts reported by JWST. Gross–Pitaevskii simulations of topological vortex halos show that defect anchoring stabilizes long-lived halo-like structures: a vortex halo retains $2.44\times$ more kinetic energy and $1.31\times$ more spatial structure (σ^2) than a defect-free clump evolving under identical conditions. Macroscopic topological defects obey the Gauss linking integral $\mathrm{Lk}(\gamma_i, \gamma_j) \in \mathbb{Z}$, suggesting effectively collisionless propagation through baryonic plasma—a topological analogue of the 150 kpc baryon–lensing offset in 1E 0657-558. Because mass requires defect seeds, the framework predicts strong suppression of baryon-free dark-matter subhalos, a falsifiable signature testable by Euclid, Roman, and LSST. All code and analysis scripts are publicly available.

INTRODUCTION

Standard ΛCDM requires collisionless particle dark matter to form halos and match large-scale structure. We explore whether the same superfluid condensate proposed in companion Letters as a model for emergent inertia and effective gravity could also address the JWST early-galaxy overabundance and the Bullet-Cluster offset, while offering a falsifiable prediction regarding baryon-free subhalos.

LOWERED COLLAPSE THRESHOLD AND THE JWST ANOMALY

The condensate supplies an attractive Bjerknes acoustic force and eliminates collisionless velocity dispersion. Solving the spherical top-hat collapse equation with the additional acoustic term

$$\ddot{\delta} + 2H\dot{\delta} = 4\pi G\rho_m\delta + \frac{4\pi G}{c_s^2}\langle\delta P_{\rm acoustic}\rangle$$

yields $\delta_c^{\rm UHF} = 1.15$. The Press-Schechter cumulative number density becomes

$$n(>M, z) = \frac{\bar{\rho}}{M}\,\mathrm{erfc}\left(\frac{\delta_c^{\rm eff}(z)}{\sqrt{2}\,\sigma(M,z)}\right).$$

At $z = 10$ and $M = 10^{10.5}M_\odot$ ($\sigma = 0.71$), the ratio relative to ΛCDM ($\delta_c^{\rm GR} = 1.686$) is exactly

$$\frac{n_{\rm UHF}}{n_{\Lambda\rm CDM}} = \frac{\mathrm{erfc}(1.15/\sqrt{2}\cdot 0.71)}{\mathrm{erfc}(1.686/\sqrt{2}\cdot 0.71)} \approx 6.01.$$

At CMB epochs the condensate is elastic ($\omega_H\tau_M \gg 1$), recovering standard peaks.

TOPOLOGICAL COLLISIONLESSNESS AND THE BULLET CLUSTER

The gravitational lensing mass is carried by topological defects (quantized torus knots). These obey the Gauss linking integral

$$\mathrm{Lk}(\gamma_i, \gamma_j) = \frac{1}{4\pi}\oint_{\gamma_i}\oint_{\gamma_j}\frac{(\mathbf{r}_i - \mathbf{r}_j)\cdot(d\mathbf{r}_i \times d\mathbf{r}_j)}{|\mathbf{r}_i - \mathbf{r}_j|^3} \in \mathbb{Z},$$

an integer topological invariant conserved under smooth deformations. Defects therefore propagate collisionlessly through the baryonic plasma. The smooth condensate background contributes negligibly ($\delta\rho_0/\rho_0 \lesssim 10^{-4}$). Consequently the lensing map remains centered on the defect distribution, reproducing the observed 150 kpc offset in 1E 0657-558 without fine-tuning.

HEURISTIC PREDICTION: BARYON-FREE SUBHALO SUPPRESSION

If inertial mass requires non-zero local condensate density ($m = C\rho_0 V$), then free-streaming or baryon-free clustering would be strongly suppressed. Defects would not form independently of baryonic seeds. The subhalo mass function for baryon-free objects would be negligible above the seeding scale ($\sim 10^8$–$10^9 M_\odot$). This prediction is falsifiable by Euclid, Roman, and LSST void-lensing

stacking and subhalo counting: confirmed baryon-free subhalos or void-lensing excess $> 3\sigma$ would rule out this scenario.

CONCLUSION

Topological defect anchoring stabilizes long-lived halo-like structures in UHF/GPE simulations, with vortex halos retaining $2.44\times$ more kinetic energy and $1.31\times$ more spatial structure than defect-free clumps. Combined with a lowered collapse threshold ($\delta_c \approx 1.15$), these results suggest a heuristic scenario that is consistent in sign with the JWST early-galaxy excess and the Bullet-Cluster baryon–lensing offset. The predicted suppression of baryon-free subhalos is testable by forthcoming surveys. All results should be regarded as exploratory estimates within a simplified condensate model, not as replacements for ΛCDM.

Public verification code is available at github.com/amiramitai/uhf. We thank the APS for consideration.